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10th International Mathematics Assessments for Schools (2020-2021)

Upper Primary Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

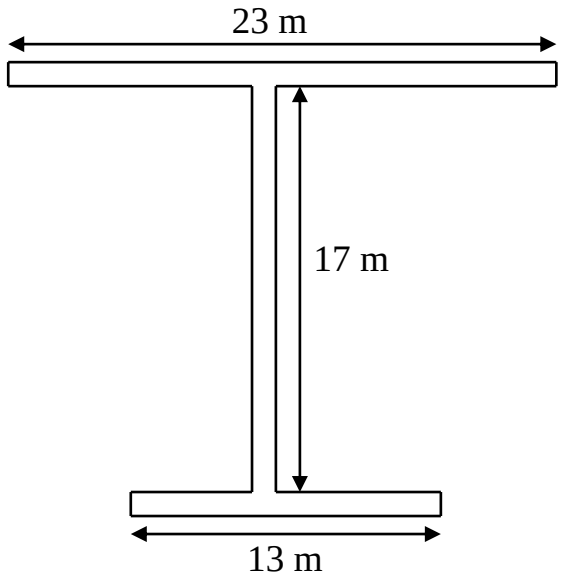
Upper Primary Division Round 2

Questions 1 to 5, 4 marks each

1. What is the integer part of the expression $6.9 + 6.99 + 6.999 + ... + 6.999999999$?
（A）60　　（B）61　　（C）62　　（D）66　　（E）69

Answer：_____

2. Jack paved the sidewalk in the garden with tiles as shown in the figure below. If the width of the sidewalk is 1 m, what is the total area, in m^2, of the sidewalk?



（A）53　　（B）104　　（C）106　　（D）221　　（E）391

Answer：_____

3. Using the dd/mm/yyyy (day/month/year) format, dates such as 01/02/2010 and 02/02/2020 are called *"Palindromic Days"* because the eight-digit numbers formed (e.g. 01022010 and 02022020) are both palindromes. What is the remainder when the *"Palindromic Day"* in the year 2021 is divided by 11?
(Note: A palindrome is a number which reads the same backward as forward.)

（A）0　　（B）1　　（C）3　　（D）7　　（E）10

Answer：_____

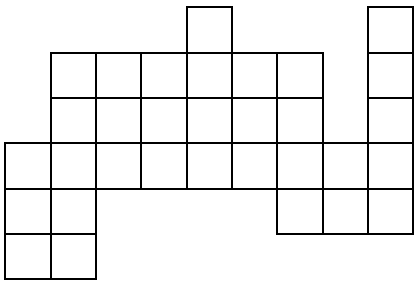
4. The table below shows the bus departure timetable of two bus routes that is shown on an electronic screen of a bus station. It can be seen that at 07:00 in the morning, both buses depart from the station at the same time for the first time. If the frequency of the departure times of both buses is fixed, when will be the third time both buses will depart at the same time?

Bus 1	07：00	07：07	07：14	07：21	07：28	…
Bus 2	07：00	07：08	07：16	07：24	07：32	…

（A）07：40　　（B）07：56　　（C）08：30
（D）08：52　　（E）09：48

Answer：_____

5. The following figure is made up of some number of identical unit squares. Now, Alex wants to form a bigger square by attaching unit squares on the figure (without changing the current figure). At least how many unit squares must he use in order to do so?



（A）32　　（B）36　　（C）49　　（D）54　　（E）81

Answer：_____

Questions 6 to 13, 5 marks each

6. Alice went to climb a mountain. Starting at the foot of the mountain, she climbed at a rate of 40 meters per minute and reached the top of the mountain in 15 minutes. On her way back down along the same path, her rate was 60 meters per minute. What is her average speed, in meters per minute, for the whole duration?

Answer：_____ meters/minute

7. There are two positive numbers such that if we divide the first number by the second number, their quotient is 3 with a remainder of 1. If the sum of both numbers is 2021, then what is the value of the first number?

Answer：_____

8. Using some number of black and white balls of the same size, we arrange the figures as shown in the diagram below. If Figure 1 uses 8 balls, Figure 2 uses 13 balls, Figure 3 uses 20 balls and so on. How many balls are there in Figure 10?

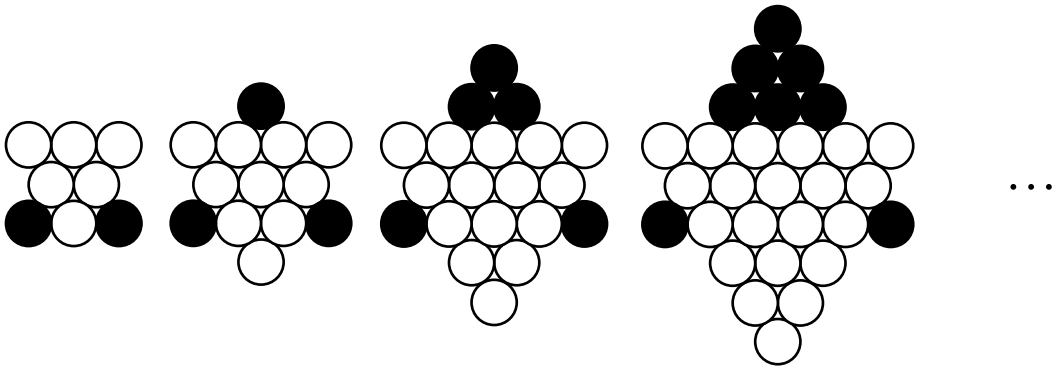
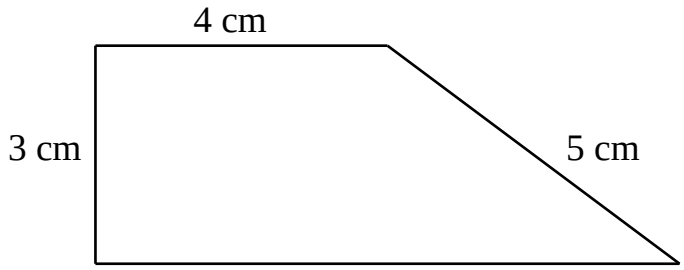


Figure 1 Figure 2 Figure 3 Figure 4

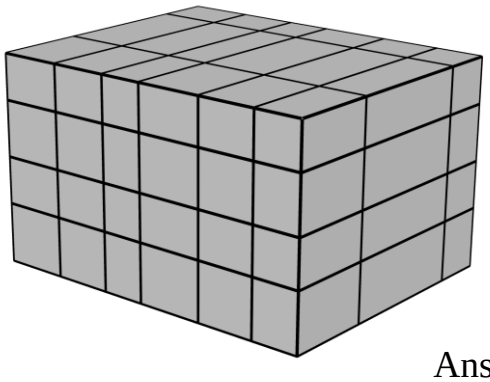
Answer : _____ balls

9. The length of three sides of a right-angled trapezoid are given as shown in the figure. What is the area, in cm^2, of this trapezoid?



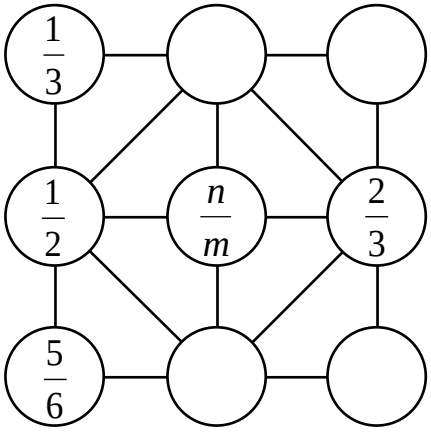
Answer : _____ cm^2

10. How many cuboids of different sizes are found in the diagram below?
(Note: if the cuboid is of the same size, but is located on a different location, then they are considered different.)



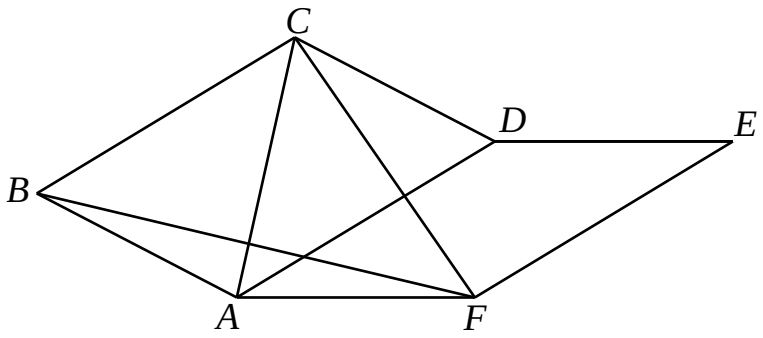
Answer : _____ cuboids

11. In the diagram, fill each of the "◯" with numbers so that the sum of the four vertices of all squares that can be found are equal. If the middle number is filled as $\frac{n}{m}$, where m and n are both positive numbers that are relatively prime. What is the value of $m+n$?



Answer : _____

12. In the figure below, *ABCD* and *ADEF* are both parallelograms and the areas of triangles *ABF* and *ACF* are 20 cm^2 and 50 cm^2, respectively. What is the area, in cm^2, of parallelogram *ADEF*?



Answer : _____ cm^2

13. In the non-terminating decimal 0.1571428..., it is known that amount the 7 digits after the decimal point already contains one complete cycle that is non-terminating and that the 202nd digit after the decimal point of this decimal is 1. Express this decimal as a fraction, in simplest terms.

Answer : _____

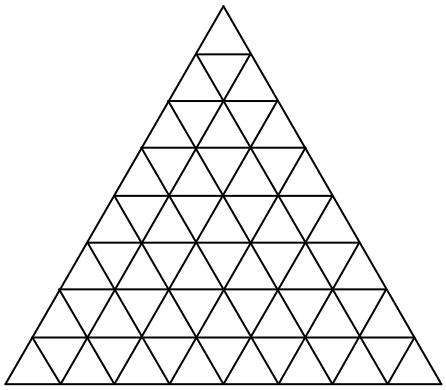
Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. Let $\overline{abcd}$ and $\overline{cdab}$ be two 4-digit numbers such that a, b, c and d are four different digits from 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. If M is the greatest common factor of both $\overline{abcd}$ and $\overline{cdab}$, then what is the largest possible value of M?

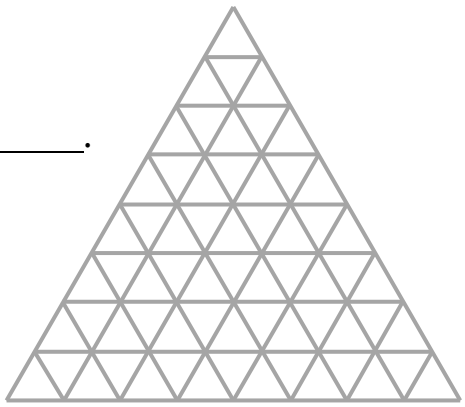
Answer : _____

15. The two figures below are composed of small equilateral triangles of the same size. We want to put several pieces of the parallelogram cardboard (which can be rotated or flipped that is shown on the left) into the equilateral triangle grid that is shown on the right without overlapping each other. The sides of these parallelogram cardboards are required to be placed along the grid lines of the equilateral triangle grid and cannot exceed the sides of it. What is the maximum number of pieces of parallelogram cardboard that can be placed into the triangular grid?
Justify your answer and show one way to place the cardboards.





The maximum number of pieces is _____.
Right figure is one way to place those cardboards.



Answer：_____